

Solution Requirements

Date	2 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The list below defines exactly what the HDI Prediction System does and how well it performs, covering both functional behaviors and non-functional quality criteria.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (what the system should do).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of the system, rather than specific behaviors (how the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system is a publicly accessible prediction tool. No user login or identity verification is required to submit a prediction request; any visitor can enter indicator values and receive a result.	Low
2	Authorization levels	Single-role, open-access application. There are no differentiated user roles (e.g., admin vs. regular user) or permission tiers — all visitors have the same access to the prediction feature.	Low
3	External interfaces	The front end communicates with a backend prediction service/API that hosts the trained machine learning model. No third-party financial services or external databases are required; the only external dependency is the hosting platform (Bolt.host).	High
4	Transactions processing	No financial transactions occur. Each user interaction is a stateless prediction request: the system accepts four indicator inputs (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita PPP), sends them to the prediction model, and returns the predicted HDI score and development tier.	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
5	Reporting	The system displays the predicted HDI score and its corresponding development tier (Very High / High / Medium / Low) directly on screen. No exportable reports, PDFs, or historical dashboards are generated in the current scope.	Medium
6	Business rules, etc.	HDI tier classification follows standard threshold bands (e.g., Very High ≥ 0.800 , High $0.700\text{--}0.799$, Medium $0.550\text{--}0.699$, Low < 0.550). Input indicator values must fall within realistic real-world ranges before a prediction is generated.	High
7	Compliance to laws or regulations	The system does not collect, store, or process any personally identifiable information — only anonymized statistical indicator values are submitted — so no GDPR/HIPAA-level data protection obligations apply.	Low
8	Other	Client-side input validation ensures all indicator values entered are numeric and within plausible ranges before being sent to the prediction model, preventing invalid or malformed requests.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction requests should return the HDI score and tier to the user without noticeable delay after submitting indicator values.	Response returned in < 2 seconds under normal load
2	Scalability	The system should continue to serve prediction requests correctly as the number of simultaneous users grows.	Supports at least 100 concurrent users without degradation
3	Security & Data Privacy	Since only non-personal statistical indicators are submitted, no sensitive personal data is collected or persisted beyond the active session.	No personally identifiable information collected or stored
4	Reliability & Availability	The hosted application should remain accessible to users with minimal unplanned downtime.	$\geq 99\%$ monthly uptime on the hosted deployment
5	Usability & Accessibility	The interface should let users enter indicator values and view results without instructions or training, on both desktop and mobile.	Clean, responsive layout usable across common screen sizes
6	Other	Maintainability: the prediction model, thresholds, and input ranges should be easy to retrain or update as new HDI data becomes available.	Model/config updates deployable without a full rebuild